HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Developed and customized core Android framework layers, including Telephony and ConnectivityService, for specialized device requirements.
- Integrated vendor-specific Radio Interface Layer (RIL) and modem firmware into the Android Open Source Project (AOSP) build.
- Optimized LTE and 5G network performance, ensuring stable connectivity and seamless handover in mission-critical communication environments.
- Debugged complex system-level issues using **ADB** and **Wireshark**, resolving latency and connectivity drop problems for enterprise devices.
- Collaborated with cross-functional teams on the **Astro Device Software** team to deliver product features, contributing to stability and quality.
- Mentored junior engineers on Android internals and best practices, fostering knowledge sharing within the development team.
- Containerized Android builds using **Docker** for consistent development and testing environments across the team.

SKILLS

Android Internals – AOSP, Telephony Framework, ConnectivityService, RIL, Modem Firmware, JNI

Languages – Java, C++

Wireless Protocols – LTE, 5G, NAS, AS, RRC

Build & Tools – Soong, Ninja, Make, ADB, Wireshark, Jira, Gerrit, Git, Gradle, QXDM, Dumpsys, Tombstone

System Performance – Memory Usage, CPU Utilization, Latency Analysis, Connectivity Optimization

Cloud & Containerization – Docker, AWS

PROJECTS

AOSP Customization for LMR Devices

- Modified and extended Android framework layers, focusing on Telephony and ConnectivityService to meet specific hardware and communication needs.
- Integrated and tested vendor RIL and modem firmware, ensuring proper communication between the OS and hardware components.
- Performed system-level debugging to resolve connectivity issues, achieving stable performance for critical LMR applications.

Android Connectivity Performance Tuning

- Fine-tuned LTE and 5G protocol stacks (NAS, AS, RRC) to enhance network stability and handover efficiency.
- Analyzed and optimized wireless communication performance using tools like QXDM and ADB, reducing connectivity drops by approximately **15%**.
- Contributed to build system configurations using Soong and Make for custom Android builds.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)